Student Marks and Grade Calculator

1. Problem Statement

Read a student's marks in three subjects, calculate the total and average, and assign a grade based on the average marks.

2. Objectives

1. Read student name and three subject marks.
2. Calculate total and average marks.
3. Use conditional statements to assign a grade.
4. Print a marks statement.

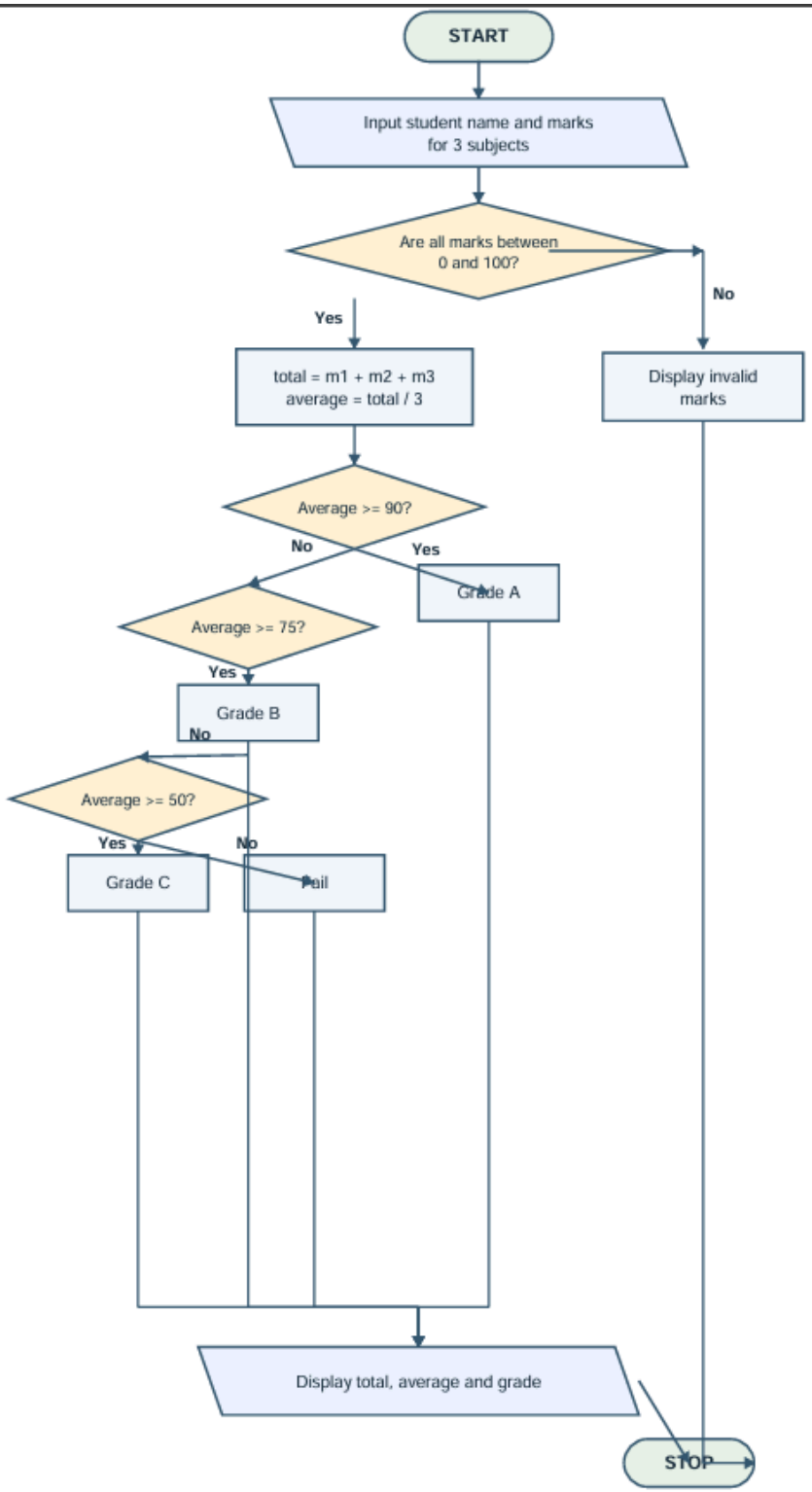
3. Assumptions

1. Each subject is out of 100 marks.
2. Marks must be between 0 and 100.
3. Grade A: 90 and above; B: 75–89; C: 50–74; Fail: below 50.

4. Algorithm

1. Start.
2. Read student name and marks in three subjects.
3. Validate that each mark is between 0 and 100.
4. Calculate total and average.
5. Use if-elif-else to assign a grade.
6. Display the marks statement.
7. Stop.

5. Flowchat


START
Input student name and marks for 3 subjects
Are all marks between 0 and 100?
No
Yes
total = m1 + m2 + m3
average = total / 3
Display invalid marks
Average >= 90?
No
Yes
Grade A
Average >= 75?
Yes
Grade B
No
Average >= 50?
Yes
No
Grade C
Fail
Display total, average and grade
STOP

6. Pseudocode

```
None
INPUT name, m1, m2, m3

total = m1 + m2 + m3
average = total / 3

IF average >= 90 THEN
    grade = "A"
ELSE IF average >= 75 THEN
    grade = "B"
ELSE IF average >= 50 THEN
    grade = "C"
ELSE
    grade = "Fail"

DISPLAY total, average, grade
```

8. Test Case and Expected Output

Input:

```
None
Name = Anu
Marks = 85, 78, 82
```

Calculation:

```
None
Total = 245
Average = 81.67
Expected grade = B
```

Program Output:

```
None
Enter student name: Anu
Enter marks in Subject 1: 85
```

```
Enter marks in Subject 2: 78
Enter marks in Subject 3: 82

----- MARKS STATEMENT -----
Student: Anu
Total: 245.0 / 300
Average: 81.67
Grade: B
```

9. Conclusion

The program calculates total and average marks and assigns a grade using if-elif-else.